Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/22/2022
PI Name:	St Hilaire, Melissa Ph.D.		
Project Title:	Towards a Real-Time Estimate of Circadian Phase During Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	msthilaire@rics.bwh.harvard.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-278-0620
Organization Name:	Brigham And Women's Hospital, Inc.		
PI Address 1:	Harvard Medical School		
PI Address 2:	221 Longwood Ave, BLI-438		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5804	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	02/01/2020	**End Date:**	01/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/31/2022 per NSSC information (Ed., 7/9/21) NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 2/3/21)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Lockley, Steven Ph.D. (Brigham And Women's Hospital, Inc.) Rahman, Shadab Ph.D. (Brigham And Women's Hospital, Inc.)		
Grant/Contract No.:	80NSSC20K0576		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The 24-hour circadian clock, located in the hypothalamus of the brain, controls the timing and pattern of sleep, alertness, performance, and mood, disruption of which has a major impact on safety. The circadian clock also controls many other aspects of human physiology, perturbation of which will affect longer-term health and well-being. It is imperative that the timing of the circadian clock is measured routinely, as a standard measure of health and behavior, during long-duration space missions to ensure that these multiple brain and body systems are not disrupted and misaligned from either each other (internal desynchronization) or with the work schedule (external desynchronization). Once measured, and if out of phase, circadian rhythms can be resynchronized to the desired time using a number of countermeasures, including bright light, melatonin, and potentially exercise and meal timing. The effectiveness of these countermeasures depends critically, however, on the circadian phase at which they are administered; mistimed countermeasures can exacerbate circadian disruption, leading to worsening of health and performance outcomes rather than improvement. Circadian phase is highly individualized, however, varying by up to 5 hours in healthy individuals under stable conditions, and is even more variable in space. To date, no feasible method exists to measure circadian phase accurately and reliably in real-time during spaceflight. Existing methods currently being used in-flight require collection of serial urine samples for up to 48 hours, from which the concentration of the urinary metabolite 6-sulphatoxymelatonin is measured to determine the timing of the circadian rhythm. Importantly, however, no technology exists to measure concentrations of urinary 6-sulphatoxymelatonin directly in-flight; therefore, knowledge about in-flight circadian disruption can be determined only during post-hoc analyses once the samples have been returned to Earth, which limits the ability to treat circadian disruption in real time. This proposal addresses this critical gap by (1) validating the use of commercially available meters to derive circadian phase in real-time from other biomarkers that exhibit robust circadian rhythms; and (2) developing a metric that estimates circadian phase from only two samples, thereby minimizing the need for serial data collection. The successful validation of this approach will enable circadian phase to be measured in real-time during spaceflight with minimal burden to the crew such that countermeasures to mitigate the effects of circadian misalignment can be tailored to individual crewmembers needs, thereby enhancing the safety, performance, and well-being of all crewmembers during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	NASA has invested significant research into the design of countermeasures to mitigate the effects of circadian misalignment. These countermeasures will be ineffective if not timed appropriately relative to circadian phase. Our real-time circadian phase estimate is expected to significantly improve the impact of such countermeasures. In addition, treatments for a wide range of conditions, including circadian rhythm sleep disorders, hypertension, and cancer, are maximally effective when timed relative to circadian phase, yet circadian phase remains absent from standard clinical assessments due to significant time and cost constraints in obtaining an accurate measure. The ability to estimate circadian phase non-invasively in real-time over a short collection window is therefore expected to enable ‘circadian medicine’ within standard clinical practice.
Task Progress:	Aim 1: Validate the use of real-time measures of saliva cortisol and blood cholesterol to derive circadian phase. To address this aim, we completed a study in 20 participants (10 males, 10 females). Participants collected three biological fluid types at home over two ~48-hour periods: saliva, blood, and urine. We collected these samples to assess several biomarkers for circadian phase using real-time assays via commercially available point-of-care devices, as well as standard assays (for validation of the real-time assays and presence of 24-hour rhythms). Our analysis focused on whether biomarkers measured from real-time devices (1) exhibited significant 24-hour rhythms during both 48-hour collection periods; and (2) circadian phase measured from these biomarker rhythms was stable across both 48-hour collection periods. In addition, for biomarkers measured via both real-time devices and via standard (i.e., non-real-time) assay methods, we compared whether rhythms and circadian phase results were similar across methods. These results are being prepared for publication. Aim 2: Develop a method to derive circadian phase from as few as three sample time points. We have developed our preliminary approach to estimate circadian phase from biomarker concentrations measured from three samples collected over a ~12-16-hour period. Due to initial delays in sample collection due to COVID-19, we started algorithm development and validation on available inpatient laboratory data, and incorporated data from our current study as results became available. We compared circadian phase estimates from this approach to the gold-standard marker of circadian phase in melatonin. These results are being prepared for publication.
Bibliography Type:	Description: (Last Updated: 12/06/2023)
Abstracts for Journals and Proceedings	Kent BA, Rahman SA, Lockley SW, St. Hilaire MA. "Towards a real-time estimate of circadian phase during spaceflight. " 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators’ Workshop, Virtual, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Kent BA, Rahman SA, Lockley SW, St. Hilaire MA. "Towards a real-time estimate of circadian phase during spaceflight. " 2021 NASA Human Research Program Investigators’ Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators’ Workshop, Virtual, February 1-4, 2021. , Feb-2021
Articles in Peer-reviewed Journals	Kent BA, Rahman SA, St Hilaire MA, Grant LK, Rüger M, Czeisler CA, Lockley SW. "Circadian lipid and hepatic protein rhythms shift with a phase response curve different than melatonin." Nat Commun. 2022 Feb 3;13(1):681. Erratum in: Nat Commun. 2022 Apr 20;13(1):2241. https://doi.org/10.1038/s41467-022-28308-6 ; PMID: 35115537; PMCID: PMC8814172 , Feb-2022
Articles in Peer-reviewed Journals	Brown LS, St Hilaire MA, McHill AW, Phillips AJK, Barger LK, Sano A, Czeisler CA, Doyle FJ 3rd, Klerman EB. "A classification approach to estimating human circadian phase under circadian alignment from actigraphy and photometry data. " J Pineal Res. 2021 Aug;71(1):e12745. https://doi.org/10.1111/jpi.12745 ; PMID: 34050968 PMCID: PMC8474125 , Aug-2021
Articles in Peer-reviewed Journals	Grant LK, St Hilaire MA, Brainard GC, Czeisler CA, Lockley SW, Rahman SA. "Endogenous circadian regulation and phase resetting of clinical metabolic biomarkers." J Pineal Res. 2021 Sep;71(2):e12752. https://doi.org/10.1111/jpi.12752 ; PMID: 34118084 , Sep-2021